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PyBispectra: A toolbox for advanced electrophysiological signal processing using the bispectrum

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Summary

Various forms of information can be extracted from neural timeseries data. Of this information, phase-amplitude coupling, time delays, and non-sinusoidal waveshape characteristics are of great interest, providing crucial insights into neuronal function and dysfunction. However, the methods commonly used for analysing these features possess notable limitations. Recent work has revealed the bispectrum – the Fourier transform of the third order moment – to be a powerful tool for the analysis of electrophysiology data, overcoming many of these limitations. Here we present `PyBispectra`, a package for bispectral analyses of electrophysiology data including phase-amplitude coupling, time delays, and non-sinusoidal waveshape.

Statement of need

Analysis of phase-amplitude coupling, time delays, and non-sinusoidal waveshape provide important insights into interneuronal communication (Canolty & Knight, 2010; Sherman et al., 2016; Silchenko et al., 2010). Studies of these features in neural data have been used to investigate core nervous system functions such as movement and memory, including their perturbation in disease (Bazzigaluppi et al., 2018; Binns et al., 2024; Cole et al., 2017; De Hemptinne et al., 2013). However, traditional methods for analysing this information have critical limitations that hinder their utility. In contrast, the bispectrum - the Fourier transform of the third order moment (Nikias & Raghuveer, 1987) - can be used for the analysis of phase-amplitude coupling (Zandvoort & Nolte, 2021), non-sinusoidal waveshape (Bartz et al., 2019), and time delays (Nikias & Pan, 1988), overcoming many of the limitations associated with traditional methods.

Despite these benefits, the bispectrum has seen relatively little use in the field of neuroscience, in part due to the lack of an accessible, easy-to-use toolbox tailored to electrophysiology data. Code written in MATLAB exists for some analyses (see e.g., github.com/sccn/roiconnect, github.com/ZuseDre1/AnalyzingWaveshapeWithBicoherence), however it is spread across multiple repositories, and often not in the form of a toolbox. Furthermore, use of this code requires a paid MATLAB license, limiting its accessibility. Code for computing the bispectrum does exist in the free-to-use Python language - e.g., Bachetti et al. (2024) - however these implementations are not tailored for use with electrophysiology data. The `PyBispectra` package

addresses these problems by providing a single, comprehensive toolbox for bispectral analysis of electrophysiology data (Figure 1), including tutorials to facilitate an understanding of these analyses in the context of neuroscience research.

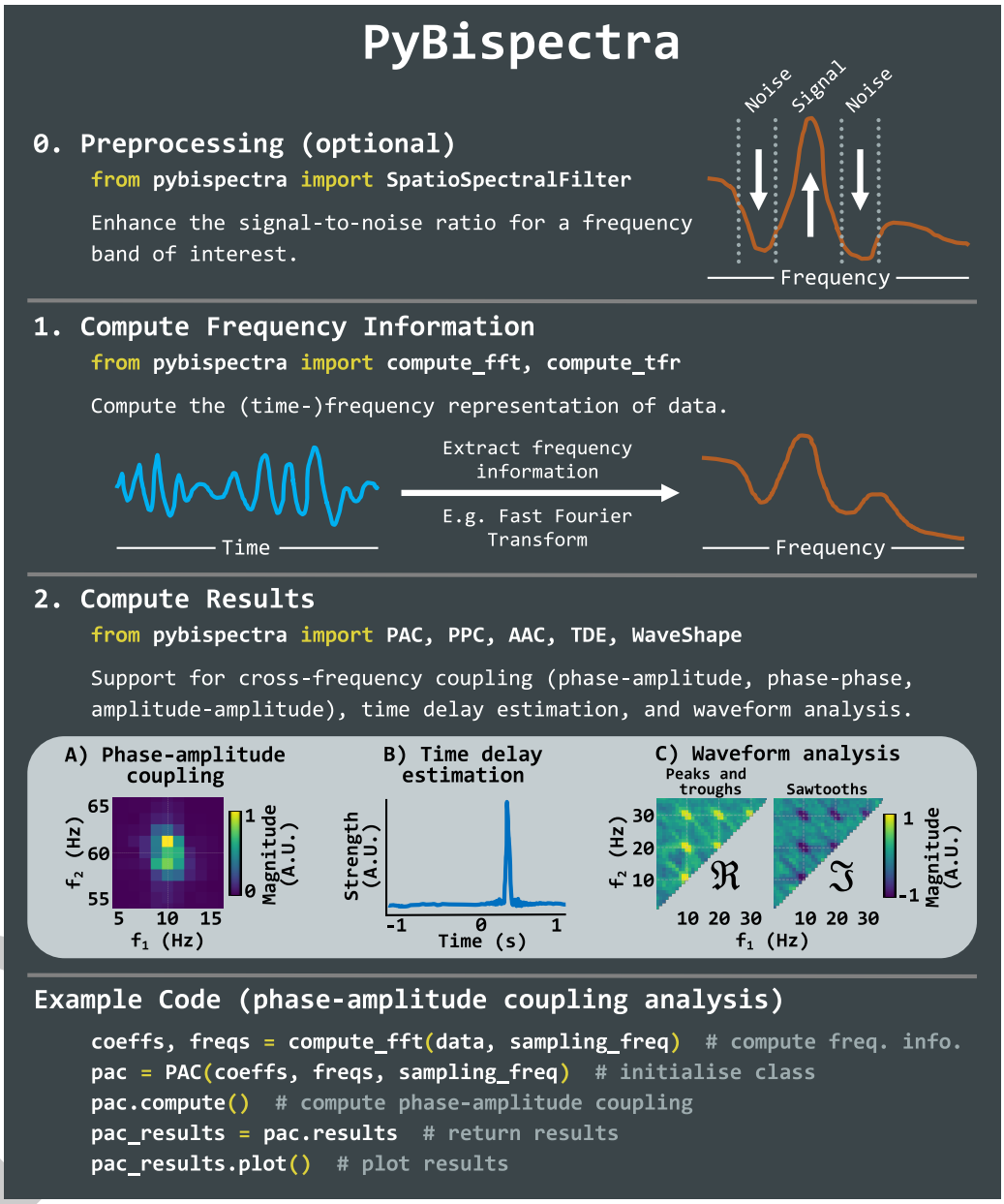

Figure 1: Overview of the `PyBispectra` toolbox. Optional preprocessing methods are supported for the multivariate analysis of waveshape. Tools are provided for computing spectral representations of timeseries data. Tools are provided for computing cross-frequency coupling, time delays, and non-sinusoidal waveshape, with schematic visualisations of results shown. Also shown is an example code snippet for analysing phase-amplitude coupling.

Features

Phase-amplitude coupling

Phase-amplitude coupling refers to the interaction between the phase of a lower frequency oscillation and the amplitude of a higher frequency oscillation. It has been posited as a mechanism for the integration of neural information across spatiotemporal scales (Canolty & Knight, 2010), with perturbations seen in disease (Bazzigaluppi et al., 2018; De Hemptinne et al., 2013). Common methods for quantifying phase-amplitude coupling include the Canolty approach (Canolty et al., 2006) and modulation index (Tort et al., 2010), which involve bandpass filtering signals in the frequency bands of interest and using the Hilbert transform to extract the phase and amplitude information. This approach has several limitations. First, the bandpass filters require precise properties that are not readily apparent, with poorly designed filters smearing information across a broad spectral range, hindering the analysis of frequency-specific interactions (Zandvoort & Nolte, 2021). Second, the Hilbert transform is a relatively demanding procedure, contributing to a high computational cost of the analysis. Finally, when analysing interactions between different signals, spurious coupling estimates can arise due to interactions within each signal which cannot be corrected for (Pellegrini et al., 2023). In contrast, the bispectrum captures phase-amplitude coupling whilst overcoming these limitations. Specifically, analysing coupling does not require bandpass filtering, preserving the spectral resolution and reducing the risk of misinterpreting results (Zandvoort & Nolte, 2021). Furthermore, bispectral analysis is highly efficient, relying on the computationally cheap Fourier transform. Finally, spurious across-signal coupling estimates can be corrected for using bispectral antisymmetrisation (Chella et al., 2014; Pellegrini et al., 2023). `PyBispectra` provides the tools for performing bispectral phase-amplitude coupling analysis, with options for antisymmetrisation and a univariate normalisation procedure that bounds coupling scores in a standardised range for improved interpretability (Shahbazi et al., 2014).

Time delays

Time delay analysis identifies latencies of information transfer between signals, providing additional insight into the physical connections between brain regions from electrophysiology data (Binns et al., 2024; Silchenko et al., 2010). A traditional time delay analysis method is cross-correlation, quantifying the similarity of signals at a set of time lags. However, this approach has a limited robustness to noise (Nikias & Pan, 1988) and a vulnerability to spurious zero time lag interactions arising due to volume conduction and source mixing in the sensor space. On the other hand, the bispectrum is resilient to Gaussian noise sources in the data (Nikias & Pan, 1988), and antisymmetrisation can be used to correct for spurious zero time lag interactions (Chella et al., 2014). `PyBispectra` provides tools for bispectral time delay analysis, with options for antisymmetrisation.

Non-sinusoidal waveshape

Non-sinusoidal signals indicate particular properties of interneuronal communication (Sherman et al., 2016), with perturbations seen in disease (Cole et al., 2017). Various non-sinusoidal properties can be identified, among them sawtooth signals and a dominance of peaks or troughs. Analysis of these characteristics can be performed on timeseries data using peak finding-based procedures - see e.g., Cole et al. (2017) - however this is computationally demanding for high sampling rate data. Furthermore, the analysis is complicated by a desire to isolate frequency-specific neural activity, with bandpass filtering suppressing non-sinusoidal information (Bartz et al., 2019). Attempts to address this remain limited by a risk of contamination from frequencies outside the band of interest - see e.g., Cole et al. (2017). In contrast, the bispectrum captures frequency-resolved non-sinusoidal information directly (Bartz et al., 2019), in a computationally efficient manner that minimises the impact of high sampling rate data. `PyBispectra` provides tools for analysing non-sinusoidal waveshape using the bispectrum, including the option of

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univariate normalisation to bound values in a standardised range for improved interpretability (Shahbazi et al., 2014).

Supplementary features

Two common issues faced in the analysis of electrophysiology data are dealing with a limited signal-to-noise ratio, and obtaining interpretable results from high-dimensional data (Cohen, 2022). Spatio-spectral decomposition is a multivariate technique that addresses these problems, capturing the key aspects of frequency band-specific information in a high signal-to-noise ratio, low-dimensional space (Nikulin et al., 2011). This decomposition can be applied for the multivariate analysis of non-sinusoidal waveshape, with extensions like harmonic power maximisation more specifically targetting non-sinusoidal information (Bartz et al., 2019). `PyBispectra` provides tools for using spatio-spectral filtering to enhance the signal-to-noise ratio and reduce the dimensionality of electrophysiology data.

Other practical features of `PyBispectra` include dedicated plotting tools for the visualisation of results to aid interpretability. Additionally, low-level compilation with Numba (Lam et al., 2015) and parallel processing are supported to improve efficiency. Generally, data formats follow conventions from popular signal processing packages like `MNE-Python` (Gramfort et al., 2013), and helper functions are provided as wrappers around `MNE-Python` and `SciPy` (Virtanen et al., 2020) tools to facilitate data processing prior to bispectral analyses. Furthermore, cross-frequency tools for amplitude-amplitude and phase-phase coupling are also provided, following literature recommendations for identifying genuine phase-amplitude coupling (Giehl et al., 2021). Finally, the various analyses are accompanied by detailed tutorials, facilitating an understanding of how the bispectrum can be used to analyse electrophysiology data.

Conclusion

Altogether, the bispectrum is a robust and computationally efficient tool for the analysis of phase-amplitude coupling, time delays, and non-sinusoidal waveshape. Bispectral approaches overcome key limitations of traditional methods, which have until now hindered neuroscience research. To aid in the uptake of bispectral methods, `PyBispectra` provides access to these tools in a comprehensive, easy-to-use package, tailored for use with electrophysiology data.

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